



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Multiple Regression and Dummy Variables

WEEK 1 · DAY 2

Created by **Jeff Lenamon**

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From one slope to a step for every rating

57.7 bp

test MAE, yesterday's line on rating score

35.5 bp

test MAE, rating dummies and duration,
today

Today

Two features, then a rating as columns: one per
level, one level left out and named.

821 bonds of invented issuers. Their spreads come from the course's own script: rating, maturity, an issuer effect, noise. Sector plays no part.

Every rating coefficient today is a difference from a reference level. Name it, or the number means nothing.

More than one feature: score and duration

```
LinearRegression().fit(train[["score", "duration"]], y_train)
```

Feature	Coefficient	Read as, other feature fixed		Training R2	Adjusted
score	39.1432 bp	per notch of rating score	score	0.5227	0.5220
duration	0.8690 bp	per year of duration	+ duration	0.5230	0.5214

Test MAE 57.7 to 57.5 bp. Training R-squared always rises with a column; adjusted R-squared charges for it.

Holding the others fixed, in this data: a statement about the fitted spread, not about cause.

A category as numbers: dummy variables

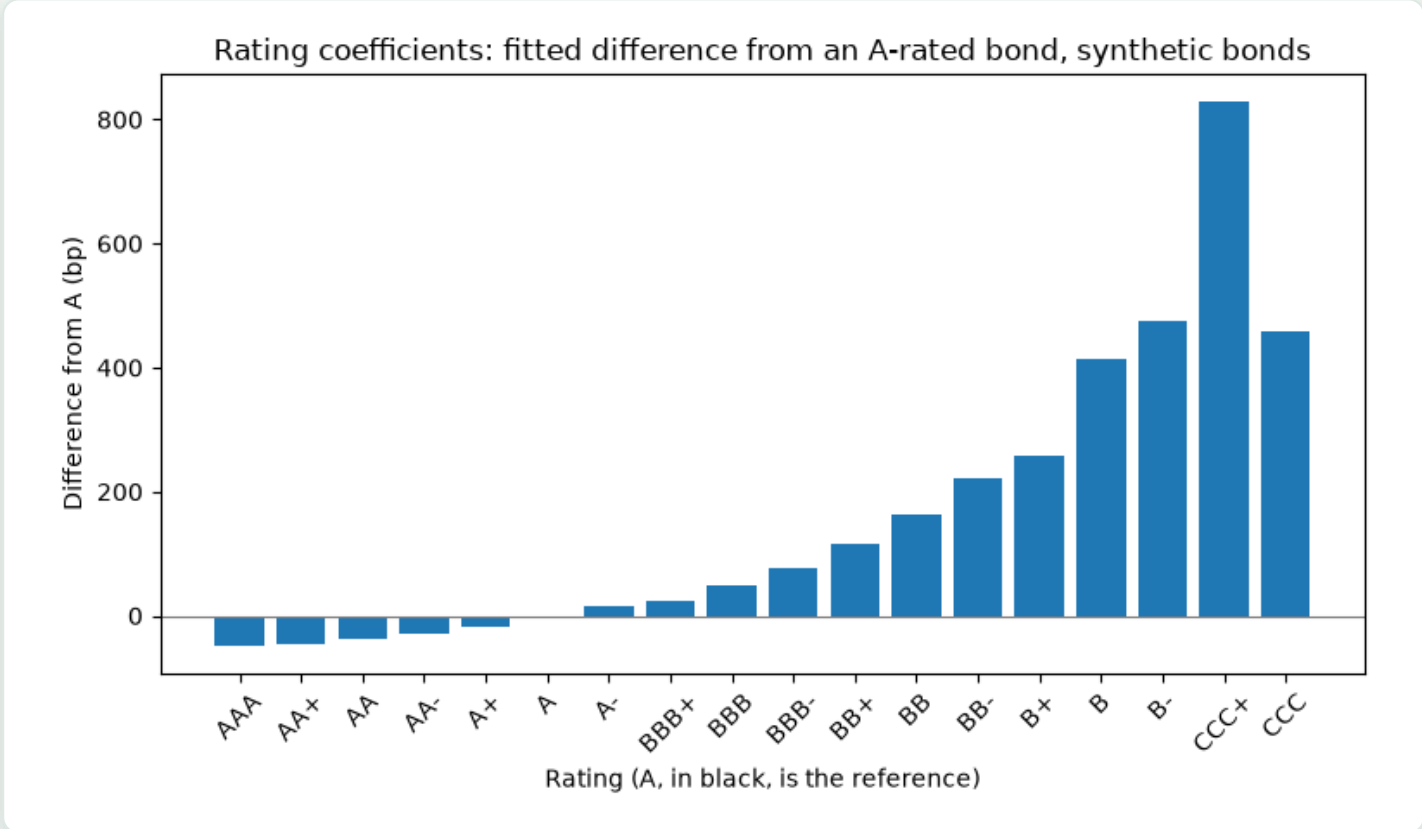
Bond	rating	rating_AAA	rating_BBB
1	AAA	1.0	0.0
2	A	0.0	0.0
3	BBB	0.0	1.0
4	A	0.0	0.0

The reference level: A

No column of its own: all zeros. The intercept stands for it, and every rating coefficient is a difference from an A-rated bond.

Which level is the reference is a choice. The fitted spreads do not depend on it; the coefficients do.

rating_dummies, with reference A



```
rating_dummies(r, levels, "A")
```

Rating	From A (bp)	Bonds
AAA	-49.0	11
BBB	48.0	42
BB	164.4	35
CCC+	828.7	9
CCC	458.1	3

Test MAE 35.5 bp, against 57.5 bp with the score.

A fitted spread, a difference from A, a residual. Never a verdict on a bond.

get_dummies(drop_first=True) drops A, not AAA

```
pd.get_dummies(bench["rating"], prefix="rating", drop_first=True)  
levels sorted as text: ['A', 'A+', 'A-', 'AA', 'AA+'] ...
```

All bonds

Leaves out A: the course's reference, by luck.

High yield bonds

Leaves out B. Nothing in the code says so.

Train and test

17 columns against 14: no test bond is AA+, B-, CCC.

rating_dummies: levels from ratings.csv, the reference named, the same columns on every set of rows.

The dummy trap: every level plus an intercept

With a column for all 18 levels, the rating columns add up to 1.0 on every row: the intercept's column, twice.

20 / 19

design columns / independent columns

-161.9 bp

coefficient on rating_A: a number with no reference

164.4 bp

rating_BB minus rating_A; the reference-A model: 164.4

Excel's LINEST on the same columns: no error; rating_CCC set to 0 with standard error 0, so every coefficient became a difference from CCC.

No error and the same fitted spreads. Only differences between rating coefficients mean anything.

Sector dummies: a feature the generator never used

Sector	From Consumer (bp)
Energy	-6.6
Financials	-8.7
Health Care	-8.9
Industrials	-7.9
Materials	-32.1

Sector	From Consumer (bp)
Technology	-13.9
Telecom	-33.5
Transportation	-33.3
Utilities	1.8

True value of every one: zero. Test MAE 36.3 bp with them, 35.5 bp without.

Nine columns fitted to noise in the training rows carry it to the test rows. Day 3 asks if they differ from zero.

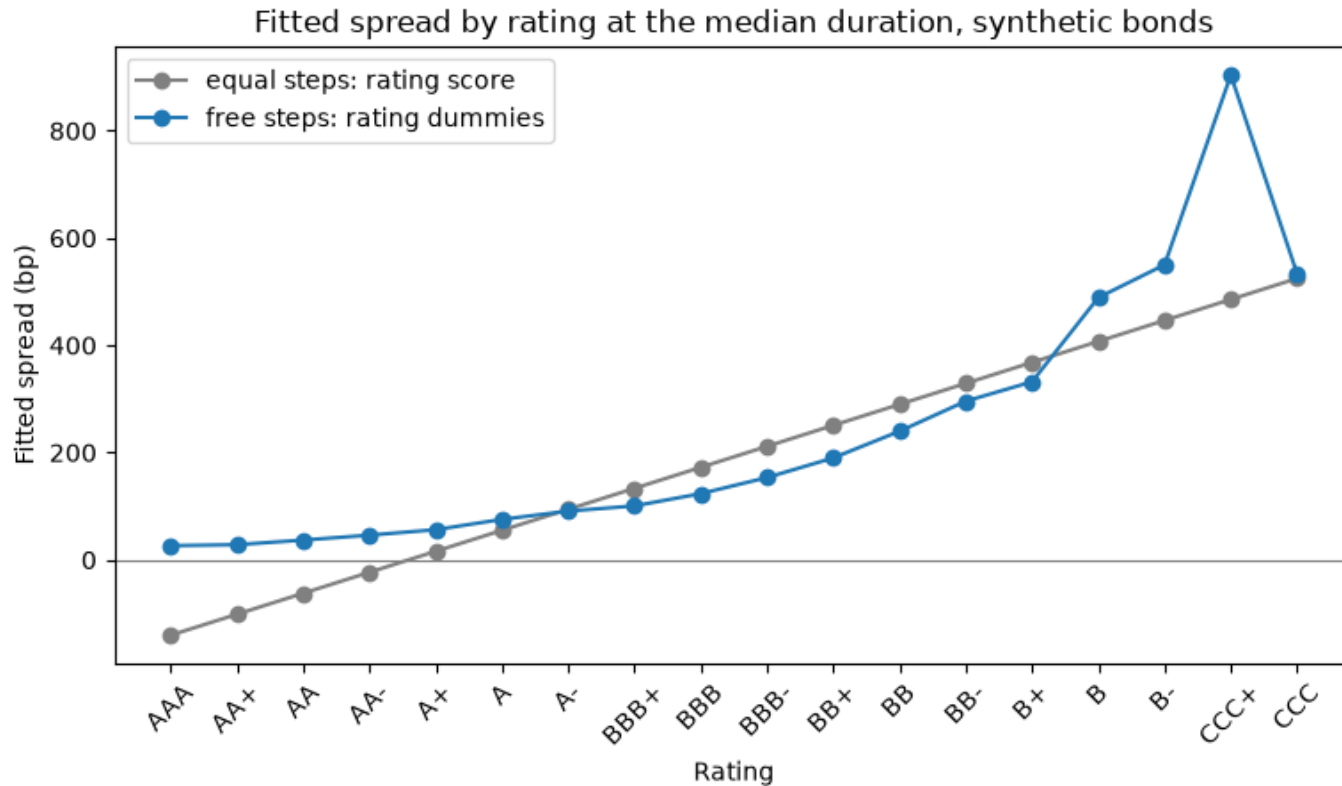
Adjusted R-squared: a charge for each feature

Model	Features	Training R2	Adjusted	Test MAE (bp)
score	1	0.5227	0.5220	57.7
score, duration	2	0.5230	0.5214	57.5
rating, duration	18	0.6727	0.6628	35.5
+ sector	27	0.6768	0.6619	36.3

$$\text{adjusted R2} = 1 - (1 - \text{R2}) \times (n - 1) / (n - k - 1)$$

It falls twice, and the test MAE agrees. Still an in-sample number: the course chooses on unseen rows.

Equal steps against free steps



Equal steps: one slope

One coefficient on all 615 bonds. Below zero at the top; test MAE 57.5 bp.

Free steps: 17 coefficients

Each on its own level's bonds; test MAE 35.5 bp.

Thin levels wander: CCC (3 bonds) sits below CCC+. At 5.88 years' duration.

Excel side by side

Excel side by side (tested in Excel)

`=--($D2="AAA")` (G2, filled down)

`=LINEST(A2:A616,B2:C616,TRUE,TRUE)`

`=LINEST(A2:A616,G2:X616,TRUE,TRUE)`

`=1-(1-INDEX(LINEST(...),3,1))*(ROWS(y)-1)/(ROWS(y)-k-1)`

`=TREND(train!y, train!X, G2:X2)` (test, filled down)

`=AVERAGE(ABS(A2:A207-AI2:AI207))`

1 for AAA, 0 otherwise

0.8690 | 39.1432 | -185.0334

R-squared 0.6727

adjusted 0.6628

the fitted spread

test MAE 35.5 bp

LINEST's first row runs backwards: duration, score, then the intercept. Read it from the right.

Excel has no dummy function: each column is a formula. TREND is predict written as a formula.

Save the day: the experiment log moves

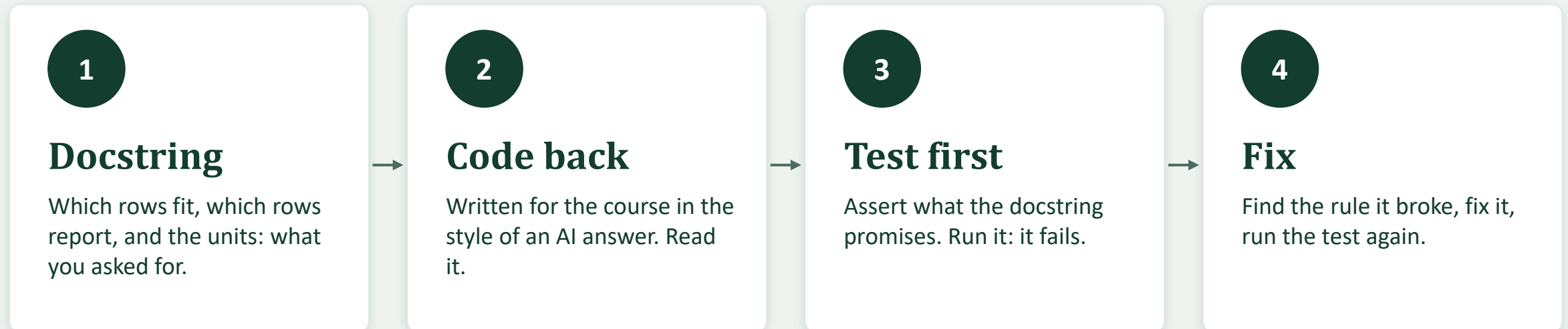
This morning: yesterday's module and log committed. Tonight: today's code, then git diff results.csv.

```
model,features,train_rows,test_rows,test_r2,test_mae_bp,test_rmse_bp
oas_on_score_line,score,615,206,0.5529,57.6620,108.3312
+oas_on_score_duration,score+duration,615,206,0.5529,57.4884,108.3237
+oas_on_rating_duration,rating(ref A)+duration,615,206,0.8179,35.5395,69.1351
+oas_on_rating_duration_sector,rating(ref A)+duration+sector(ref Consumer),615,206,0.8149,36.3359,69.7107
```

One context line, three added lines: the test MAE column reads the day.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: `get_dummies(drop_first=True)` drops A, but the summary calls every coefficient "the spread over AAA".

Code that runs is not the same as code that is right.

What you can do now

- Each coefficient is read with the other features fixed
- A category becomes columns; one level is the named reference
- `drop_first` leaves out whatever sorts first
- The dummy trap runs without an error
- Adjusted R-squared charges; the test rows decide

Week 1, Day 3 Coefficients, p-values, and multicollinearity: statsmodels and sector's p-values